Fiscal Year:	FY 2010	**Task Last Updated:**	FY 02/02/2011
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Methods for the Assessment of Gastrointestinal Physiology and Function in a Lunar Analog Environment		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	08/11/2008	**End Date:**	09/30/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Goodwin, Thomas	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2010 (from 8/10/2010) per C. Guidry/JSC (06/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Boyd, Jason (USRA) Vaksman, Zalman (Wyle Labs)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Earlier studies of gastrointestinal (GI) function in space indicate an increase in GI transit time and a significant reduction in the drug/nutrient absorption. Technical difficulties associated with conducting such studies have been in-part responsible for the inadequate research in this area. The Simulated Lunar Environment Project provides a vehicle to assess the effects of adaptation to reduced gravity. The Lactulose Breath Hydrogen test (LBHT), an indirect measure of GI motility has primarily been used due to its un-invasive. However, limitation, including accuracy and specificity, present the need for a more direct test. New capsule endoscopy technology provides a direct measurement of GITT as well as GI pH, monomotry and temperature. The effects of altered GI function on the behavior of ingested content has also never been studied. Disposition profiles of drugs like acetaminophen (APAP) serve as effective markers for gastric emptying and absorption. Although previously saliva was used to predict drug plasma concentrations this procedure has never been fully validated in reduced gravity. Therefore we propose to evaluate influence of reduced gravity on GI function using the SmartPill® technology. Further, these studies will validate the use of saliva sampling to predict plasma concentrations and evaluate the absorption of APAP in solid, liquid and gel-cap formulations. Objectives The overall goal is to examine the effects of microgravity analog environments on GI physiology. Specific Objective: Conduct a feasibility study to examine and validate new technology, SmartPill® as a tool to assess the effect of ABR and LBA on GI physiology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Validation of noninvasive diagnostic technology for GI physiopathology associated with space flight environmental conditions. Identification of prospective changes in GI function leading to a reduction in drug absorption associated with microgravity (BR risk #s 17 and 19).
Task Progress:	Eight subjects completed the protocol once before, two times during and once after HDT bedrest according to the protocol. All pertinent GI physiology variables data was compiled using vendor provided software. Analysis of results on relevant GI variables of motility, pH, and pressure revealed that there is significant inter- and intra-subject variability for these variables. Given the small N and data variability, it is concluded that existing data are insufficient for validation of the device for noninvasive assessment of GI physiological parameters in healthy, normal subjects.
Bibliography Type:	Description: (Last Updated: 06/23/2025)